

Gabriel Geffen

[Website](#) [Google Scholar](#) [LinkedIn](#)

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EDUCATION

University of Arizona

Doctor of Philosophy in Civil Engineering

Dissertation: Emerging Technologies for Resilient Highway Operations:

Implications for Traffic Predictions and Socially Compatible Driving

Advisor: [Dr. Yao-Jan Wu, P.E.](#)

Minor: Management Information Systems (Advisor: [Dr. Agrim Sachdeva](#))

Tucson, AZ

Exp. May 2027

University of Arizona

Bachelor of Science in Civil Engineering

Tucson, AZ

May 2023

Pima Community College

Arizona General Education Curriculum

Tucson, AZ

May 2021

RESEARCH INTERESTS

Data-Driven Transportation Systems

Traffic Control and Operations

Transportation Safety and Human Behavior

Intelligent Transportation Systems

Traffic Modeling, Forecasting, and Simulation

Machine Learning and Causal Inference in Transportation

Connected, Automated, and Electric Vehicles

Transportation Cybersecurity and System Resilience

PROFESSIONAL EXPERIENCE

University of Arizona

Lab Manager

Smart Transportation Laboratory (STL)

Department of Civil & Architectural Engineering & Mechanics

Tucson, AZ

2026–Present

University of Arizona

Research Assistant

Center for Applied Transportation Sciences (CATS)

Department of Civil & Architectural Engineering & Mechanics

Tucson, AZ

2022–Present

Psomas Inc.

Civil Engineering Intern

Land Development Team

Tucson, AZ

2021–2022

University of Arizona

Research Assistant

Turbulence and Flow Control Laboratory

Tucson, AZ

Summer 2021

Department of Aerospace and Mechanical Engineering

Pima Community College
Mathematics Tutor
 SOAR Program

Tucson, AZ
 2019–2020

HONORS AND AWARDS

National Awards and Fellowships

National Operations Center of Excellence (NOCoe) TRB Student Contest Winner	2026
TRB Freeway Operations Committee Best Young Professional/Student Paper Award	2025
Dwight David Eisenhower Transportation Fellowship Program (FHWA)	2023, 2024
National Science Foundation Graduate Research Fellowship	2023

University-Level Awards

Herbold Fellowship	2023
Lyon Family Civil Engineering Scholarship Endowment	2022, 2023
American Society of Highway Engineers Scholarship	2022
R.A. Jimenez Highway Scholarship	2022
Bill Lloyd ABA Scholarship	2022
Southern AZ Branch of ASCE Scholarship	2022
Gooding-Anderson Scholarship	2022

Professional Society Awards

Jenny Grote Leadership Award, Arizona Institute of Transportation Engineers	2026
Hank Warner Memorial Scholarship, Southern Arizona Institute of Transportation Engineers	2024, 2026
Best Graduate Research Paper, Intelligent Transportation Society of Arizona	2025
ITS Arizona Graduate Scholarship	2025
1st Place – Arizona Transportation Summit, Intelligent Transportation Society of Arizona	2025

Student Competitions and Design Awards

1st Place – Transportation Competition, ASCE Intermountain Southwest Region Symposium	2022
3rd Place – Timber-Strong Design Build, ASCE Intermountain Southwest Region Symposium	2022

Other Honors

Summa Cum Laude, University of Arizona	2023
National Science Foundation Research Experience for Undergraduates (REU)	2021

SPONSORED PROJECTS

1. SPR-788: Assessing the Consumption Costs of Motor Vehicles on Arizona Roads and Bridges
 Sponsor: Arizona Department of Transportation (ADOT)
 PI: [Dr. Tejo Bheemasetti](#)
 Total Budget: \$400,000.

Aug. 2024 - Dec. 2026

2. NCHRP 07-34: Artificial Intelligence for Transportation Systems Management and Operations Applications
Sponsor: National Cooperative Highway Research Program (NCHRP)
PI: [Dr. Yao-Jan Wu](#)
Total Budget: \$450,000. Apr. 2024 - Oct. 2026
3. SPR-792: State of the Practice of a Data Management Office Within DOTs
Sponsor: Arizona Department of Transportation (ADOT)
PI: [Dr. Mingfeng Shang](#)
Total Budget: \$55,000. Aug. 2024 - Sep. 2025
4. Impact of Transportation Electrification on Air Quality and Health in Arizona
Sponsor: University of Arizona, Technology and Research Initiative Fund (TRIF)
PI: [Dr. Chris Lim](#)
Total Budget: \$100,000. Aug. 2024 - Sep. 2025
5. Traffic Data Analytics Platform Development for TSMO:
Statewide Mobility Analytics in Real-Time (SMART) Tool Expansion
Sponsor: Arizona Department of Transportation (ADOT)
PI: [Dr. Yao-Jan Wu](#)
Total Budget: \$200,000. Aug. 2022 - July 2024
6. Traffic Data Analytics Platform Development for TSMO – Expansion of Ramp Metering Evaluation Tool
Sponsor: Arizona Department of Transportation (ADOT)
PI: [Dr. Yao-Jan Wu](#)
Total Budget: \$200,000. May 2020 - July 2022

SELECTED PROPOSAL WRITING

1. SPR-803: Enhancing Prediction of Traffic Delays Using Data-Driven Modeling Not Funded
Sponsor: Arizona Department of Transportation (ADOT)
PI: [Dr. Mingfeng Shang](#)
Co-PIs: [Dr. Edward Samglik](#), [Dr. Brendan Russo](#), [Dr. Yao-Jan Wu](#)
Total Budget: \$150,000; Duration: 16 months

PUBLICATIONS

Refereed Journal Articles (*: indicates the corresponding author)

- [J3] **Gabriel Geffen***, Pouya Jalali Khalilabadi, Bharat Pathivada, and Yao-Jan Wu. Leveraging Multisource Data for a Comprehensive Investigation of Split Failure Occurrences. *Journal of Transportation Engineering, Part A: Systems*, 152(6):04026028, 2026. [Online Access](#).
- [J2] **Gabriel Geffen**, Mingfeng Shang*, and Yao-Jan Wu. Novel Weather Severity Classifications Based on Speed Reductions Utilizing Machine Learning Techniques. *Transportmetrica A: Transport Science*, pages 1–23, 2025 [Online Access](#).
- [J1] **Gabriel Geffen***, Adrian Cottam, Henrick Haule, and Yao-Jan Wu. Evaluation of Factors Influencing Ramp Metering Compliance Using Controller Event–Based Data. *Transportation Research Record*, 2679(11):306–320, 2025. [Online Access](#).

Manuscripts Under Review (*: indicates the corresponding author)

- [JR3] Agrim Sachdeva*, **Gabriel Geffen**, Mingfeng Shang, and Yao-Jan Wu. Scenario-Aware Traffic Forecasting for Operational Decision Support: A Spatio-Temporal Foundation Model Approach. *Information Systems Research*, Under Review.

[JR2] **Gabriel Geffen**, Jun Zhao, Mingfeng Shang*, Shian Wang, and Yao-Jan Wu. Safety, Mobility, and Environmental Impacts of Driver-Assistance-Enabled Electric Vehicles: An Empirical Study. *International Journal of Sustainable Transportation*, Under Review.

[JR1] **Gabriel Geffen**, Mingfeng Shang*, and Yao-Jan Wu. From Crash to Congestion: Predicting Post-Incident Speed Recovery Using Probe Data. *ITE Journal*, Under Review.

Technical Reports

[R2] Mingfeng Shang, Yao-Jan Wu, and **Gabriel Geffen**. State of The Practice of a Data Management Office Within DOTs. 2025. [Online Access](#).

[R1] **Gabriel Geffen**, Adrian Cottam, and Yao-Jan Wu. Traffic Data Analytics Platform Development for TSMO: Statewide Mobility Analytics in Real-Time (SMART) Tool Expansion. 2024. [Online Access](#).

PRESENTATIONS

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- [P13] [Poster] **Geffen, G.**, Zhao, J., Shang, M., Wang, S., & Wu, Y.-J. (Expected June 2026)
Impacts of Driver Assistance-Enabled Electric Vehicles on Safety, Mobility, and the Environment: An Empirical Study
 ASCE International Conference on Transportation & Development (ICTD), Detroit, MI
- [P12] [Invited Talk] **Geffen, G.** (April 2026)
Novel Weather Severity Classifications Based on Speed Reductions Utilizing Machine Learning Techniques
 Southern Arizona Institute of Transportation Engineers Monthly Luncheon, Tucson, AZ
- [P11] [Student Poster Competition] **Geffen, G.** (March 2026)
Impacts of Driver Assistance-Enabled Electric Vehicles on Safety, Mobility, and the Environment: An Empirical Study
 AAA Safe Mobility Conference, Seattle, WA
- [P10] [Poster] **Geffen, G.**, Zhao, J., Shang, M., Wang, S., & Wu, Y.-J. (January 2026)
Impacts of Driver Assistance-Enabled Electric Vehicles on Safety, Mobility, and the Environment: An Empirical Study
 Transportation Research Board (TRB) Annual Meeting, Washington, DC (TRBAM-26-04121)
- [P9] [Poster] **Geffen, G.**, Shang, M., & Wu, Y.-J. (June 2025)
Systematically Assessing and Comparing Data-Driven Traffic Speed Prediction Models After Crashes
 ASCE International Conference on Transportation & Development (ICTD), Glendale, AZ
- [P8] [Poster] **Geffen, G.**, Jalali Khalilabadi, P., Pathivada, B., & Wu, Y.-J. (June 2025)
Leveraging Multisource Data for Insights into Split Failure Occurrences
 ASCE International Conference on Transportation & Development (ICTD), Glendale, AZ
- [P7] [Oral] **Geffen, G.** (April 2023)
Novel Weather Severity Classifications Based on Speed Reductions Utilizing Machine Learning Techniques
 ITS Arizona Student Transportation Summit, Virtual
- [P6] [Poster] **Geffen, G.**, Cottam, A., Haule, H., & Wu, Y.-J. (January 2025)
Evaluation of Factors Influencing Ramp Metering Compliance Using Controller Event-Based Data
 Transportation Research Board (TRB) Annual Meeting, Washington, DC (TRBAM-25-02910)
- [P5] [Poster] **Geffen, G.**, Shang, M., & Wu, Y.-J. (January 2025)
Assessing Key Factors Influencing Post-Crash Traffic Speeds with Data-Driven Approaches
 Transportation Research Board (TRB) Annual Meeting, Washington, DC (P25-21346)

- [P4] [Oral] **Geffen, G.**, & Cheshko, S. (September 2024)
SMART Teamwork: ADOT and UA Collaboration for Operations
 Conference on Roads and Streets, Tucson, AZ
- [P3] [Oral] **Geffen, G.**, & Cheshko, S. (April 2024)
Development of a Traffic Analytics Platform for Transportation Systems Management and Operations (TSMO)
 ITE/IMSA Annual Spring Conference, Glendale, AZ
- [P2] [Oral] **Geffen, G.** (November 2023)
Statewide Mobility Analytics in Real Time (SMART): A Web-Based Tool for Large-Scale Monitoring and Analysis of Roadway Networks
 International Road Federation (IRF) Global Road to Tomorrow (R2T) Conference, Phoenix, AZ
- [P1] [Oral] **Geffen, G.** (April 2023)
Developing a Statewide Web-Based Crash Speed Analysis Tool Using Crowdsourced Data
 ITS Arizona Student Transportation Summit, Virtual

TEACHING AND MENTORING EXPERIENCE

Teaching

- **Teaching Assistant:**
 - CE 363: Transportation Engineering and Pavement Design, University of Arizona Fall 2022 & Fall 2025
 - Responsibilities: Graded assignments, held weekly office hours, and led weekly in-person lab sessions

Students Mentoring

1. Udaya Adhikari, University of Arizona Aug. 2026 - Present
 – 2nd year Ph.D. student

Students Mentored

1. Mateo Pimienta, University of Arizona Jan. 2025 - Dec. 2025
 – Undergraduate Student
 – Assisted with research tasks (data collection and processing) and website administration
2. Sofia Iliev, University of Arizona Aug. 2024 - May 2025
 – Undergraduate Student
 – Current Affiliation: AECOM
 – Project Joined: State of the Practice of a Data Management Office Within DOTs
3. Ela Aldrete, University of Arizona Aug. 2024 - May 2025
 – Undergraduate Student
 – Project Joined: Impact of Transportation Electrification on Air Quality and Health in Arizona

LEADERSHIP & PROFESSIONAL SERVICE

Professional Memberships

- Student Member, American Society of Civil Engineers (ASCE)
- Student Member, Institute of Transportation Engineers (ITE)

Leadership Experience

- President, ITE Student Chapter, University of Arizona 2024 – 2026
 - Lead chapter operations, meetings, and events
 - Coordinate with faculty advisors and professional partners

- Vice President, ITE Student Chapter, University of Arizona
 - Supported chapter leadership and event planning
 - Assisted with member engagement and organization activities

2022 – 2023

Referee Service

- Journal Paper Refereed:
IEEE Transactions on Intelligent Transportation Systems, Transportmetrica A: Transport Science, Transportmetrica B: Transport Dynamics, Transportation Research Record
- Conference Paper Refereed:
Transportation Research Board Annual Meeting, IEEE Intelligent Transportation Systems Conference

LICENSES & CERTIFICATIONS

- **Engineer in Training (EIT)**, Civil – Arizona, License No. 01381